

Portfolio Optimization & Efficient Frontier Analysis (Excel)

Objective:

To construct an optimized equity portfolio using historical stock data, balancing risk and return based on **Modern Portfolio Theory (MPT)** principles. The goal was to identify the **optimal weights** of selected stocks that maximize the **Sharpe ratio** while visualizing the **efficient frontier**.

Data Used:

- Monthly stock prices of **Tata Motors, HDFC Ltd, TCS, and Apollo Hospitals** from the past 5 years.
- Calculated **monthly returns, annualized returns, and standard deviations**.

Methodology:

1. Return & Risk Calculation:

- Computed **monthly returns** for each stock.
- Annualized **expected returns** and **standard deviations**.
- Constructed the **covariance matrix** of stock returns to measure portfolio risk.

2. Simulation:

- Generated **5,000 random portfolios** using Excel to explore various combinations of weights.
- Calculated **portfolio expected return, portfolio standard deviation, and Sharpe ratio** for each simulated portfolio.

3. Optimization (Solver):

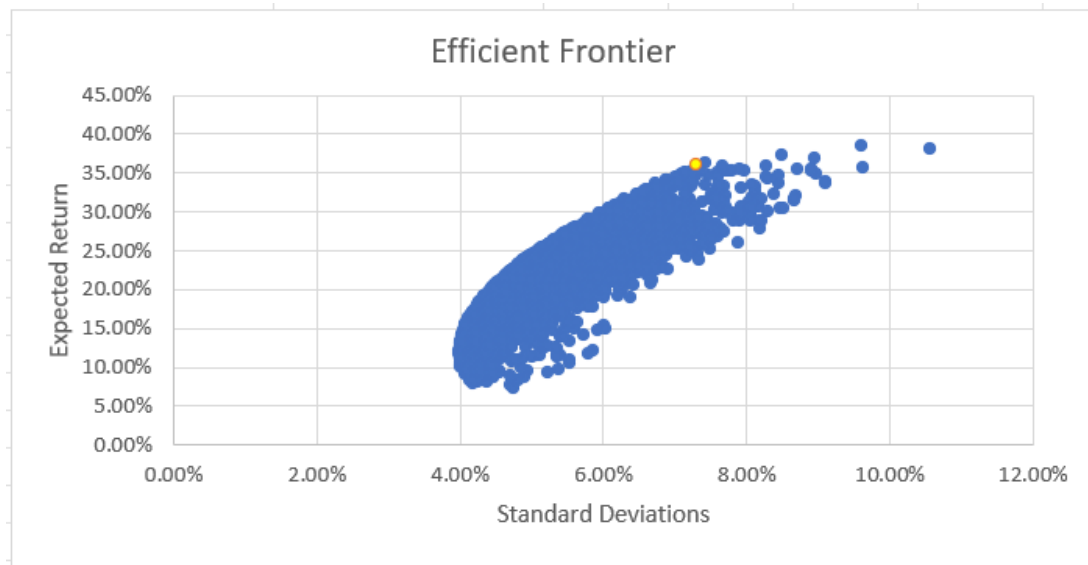
- Used Excel **Solver** to find the **optimal weights** that maximize the **Sharpe ratio**.
- Constraints applied:
 - Sum of weights = 1
 - Weights ≥ 0 (no short selling)

4. Efficient Frontier:

- Plotted the **Efficient Frontier** using simulated portfolios.
- Highlighted the **Tangency Portfolio** (maximum Sharpe ratio) on the curve for clear visualization.

Security	TATAM	HDFC	TCS	APOLLOH	Total
Annual Return	40.43%	8.34%	5.79%	33.08%	
Standard Deviation	40.97%	17.06%	21.97%	30.17%	
Weights	40.39%	0.00%	0.00%	59.61%	
Random Array	11.13%	11.22%	22.16%	95.92%	140.42%

Portfolio Summary	
Expected Return(R_p)	36.05%
Standard Deviation (SDP)	7.29%
Risk Free Rate	5.50%
Total Weight	100.00%
Sharp Ratio	4.19



Key Learning:

- Gained practical experience in **portfolio construction and diversification** by combining multiple stocks and understanding their interactions.
- Developed a deep understanding of the **risk–return trade-off** and how investors can optimize portfolios to achieve desired returns with minimum risk.
- Learned to apply **Modern Portfolio Theory (MPT)** concepts in practice, including efficient frontier construction and the tangency portfolio.
- Enhanced **quantitative and analytical skills** by calculating returns, variances, covariances, and using Excel for statistical analysis.
- Mastered **Excel modeling techniques**:
 - Simulation of 5,000 portfolios
 - Solver optimization for maximum Sharpe ratio
 - Conditional calculations and matrix-based formulas for portfolio risk
- Gained hands-on experience in **financial decision-making tools**, which are directly applicable to roles like equity research, investment analysis, risk management, and wealth management.
- Learned how to **interpret and visualize financial data effectively**, including presenting the efficient frontier and highlighting the optimum portfolio point for decision-making.
- Developed **problem-solving skills** by balancing practical constraints (weights sum to 1, no short selling) with theoretical objectives (maximizing Sharpe Ratio).